

Preclass Learning: Advanced Data Visualization (ICA 3)

1. Mastery of the Grammar of Graphics (GoG)

The GoG provides a foundation for building any graphic by combining independent layers:

- **Aesthetics (Aesthetics):** Controls how data variables are mapped to visual properties (color, size, shape, x/y position).
- **Geometries (Geoms):** The visual elements drawn to represent data (e.g., `geom_point()`, `geom_boxplot()`, `geom_sf()`).
- **Scales:** Determine the range and appearance of the aesthetic mapping (e.g., `scale_color_viridis()`, `scale_y_log10()`).
- **Facets:** Splitting the plot into multiple sub-plots based on a categorical variable (`facet_wrap()` or `facet_grid()`).

2. Customization and Refinement

- **Themes (`theme()`):** Used to control non-data elements like backgrounds, grid lines, font styles, and axis labels to improve the professional look and feel.
- **Annotation:** Adding text, lines, or shapes directly to the plot to highlight specific data points or explain trends (e.g., `annotate()` or `geom_text()`).
- **Color Theory:** Understanding how to use sequential (for continuous data), diverging (for data around a midpoint), and qualitative (for categorical data) color palettes effectively.

3. Visual Ethics

- **Data-Ink Ratio:** Maximizing the visual representation of data and minimizing non-data elements (Tufte's principle).
- **Truthfulness:** Avoiding visual distortion, such as truncated axes, that can mislead interpretation.
- **Accessibility:** Choosing color palettes and contrast levels that are accessible to colorblind viewers.